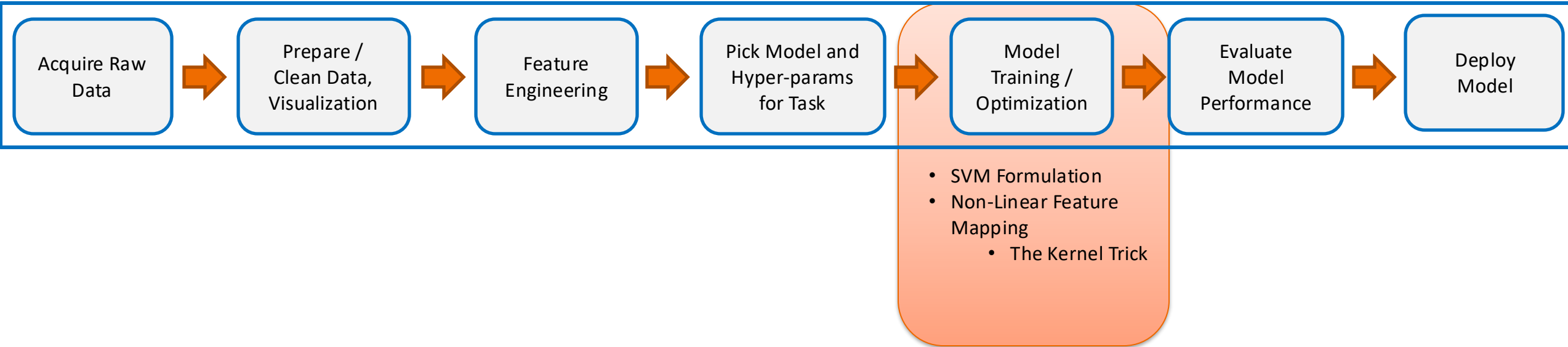


Focus for this lecture



Linear Classifier

- Decision boundary: Hyperplane

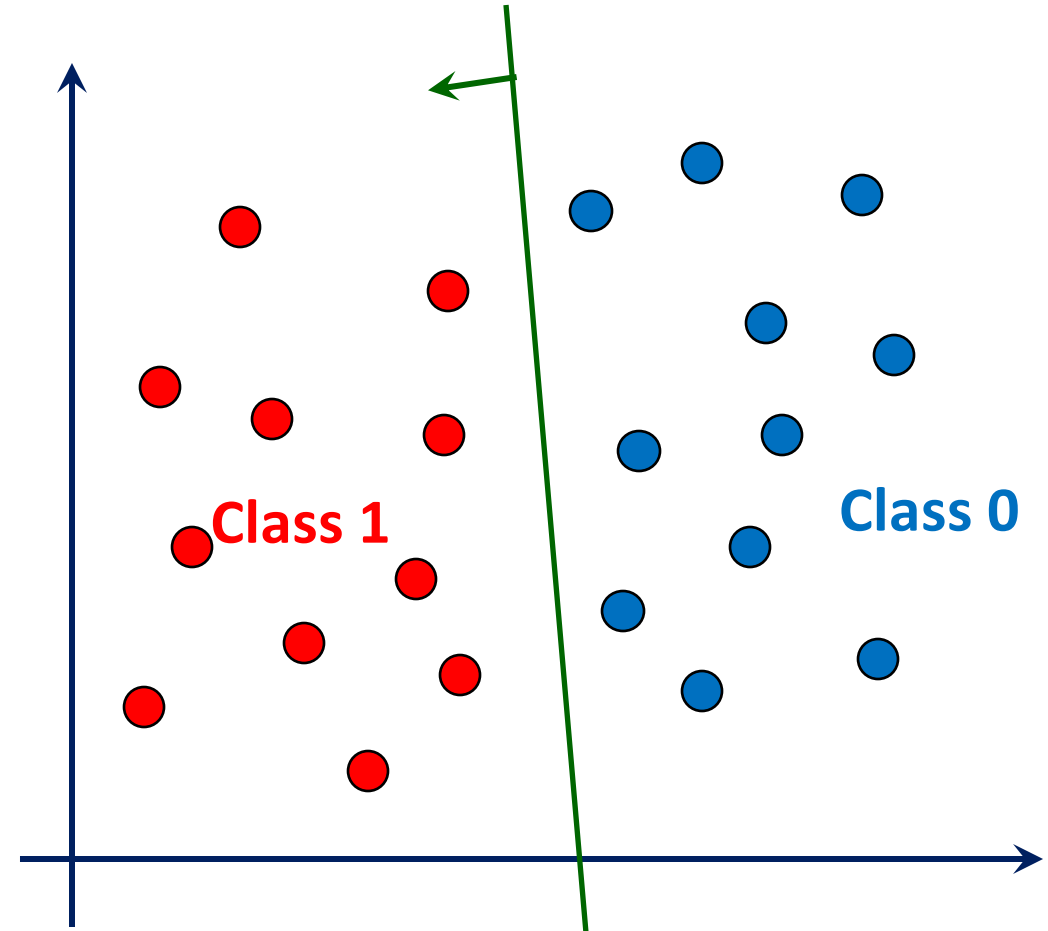
$$W^T X + b = 0$$

- Class 1 lies on the positive side

$$W^T X + b > 0$$

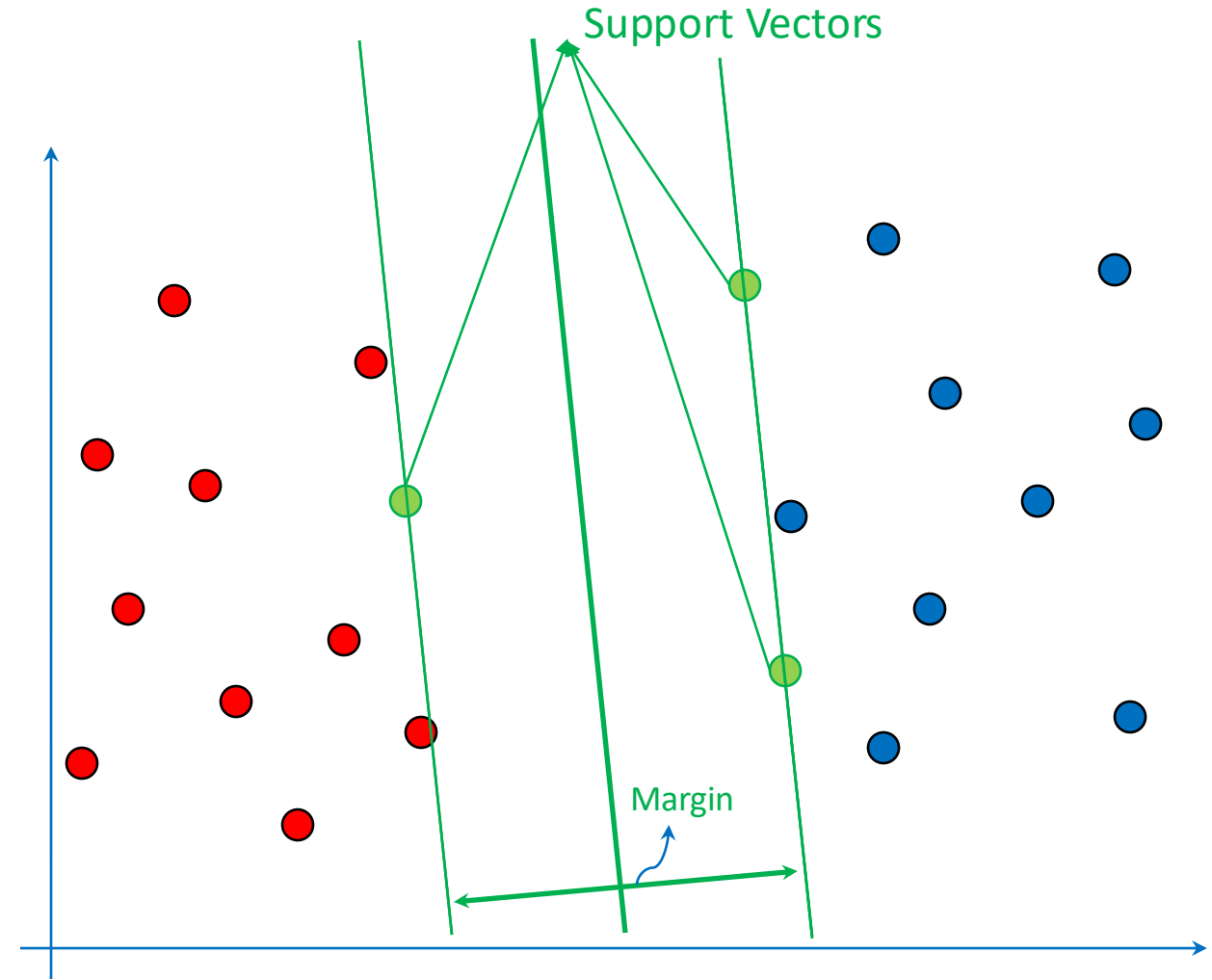
- Class 0 lies on the negative side

$$W^T X + b < 0$$



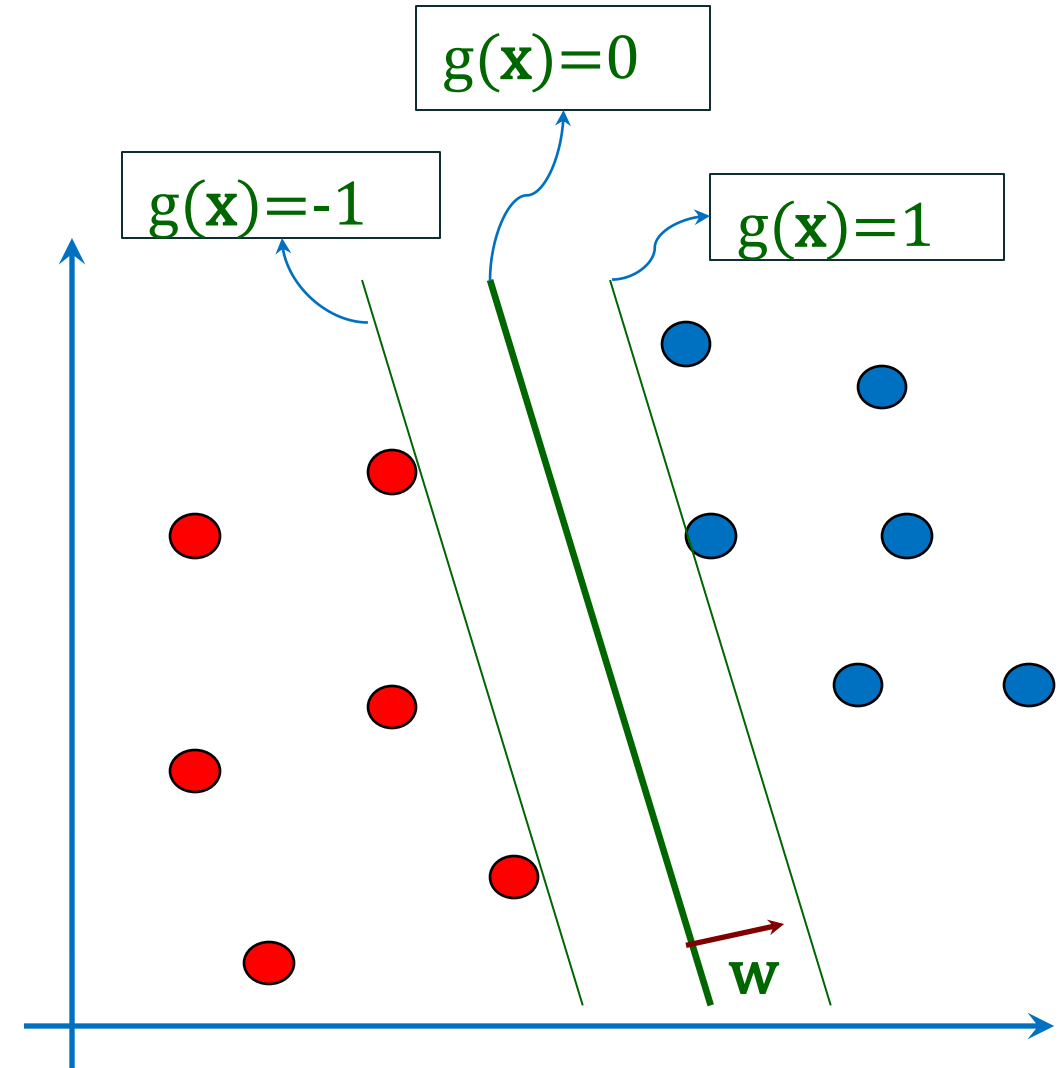
Max-Margin Classification

- A Large Margin will reduce the chance of misclassifying future test samples
- In other words, large-margin classifiers will generalize better.
- Samples at the boundary support the margin: called Support Vectors



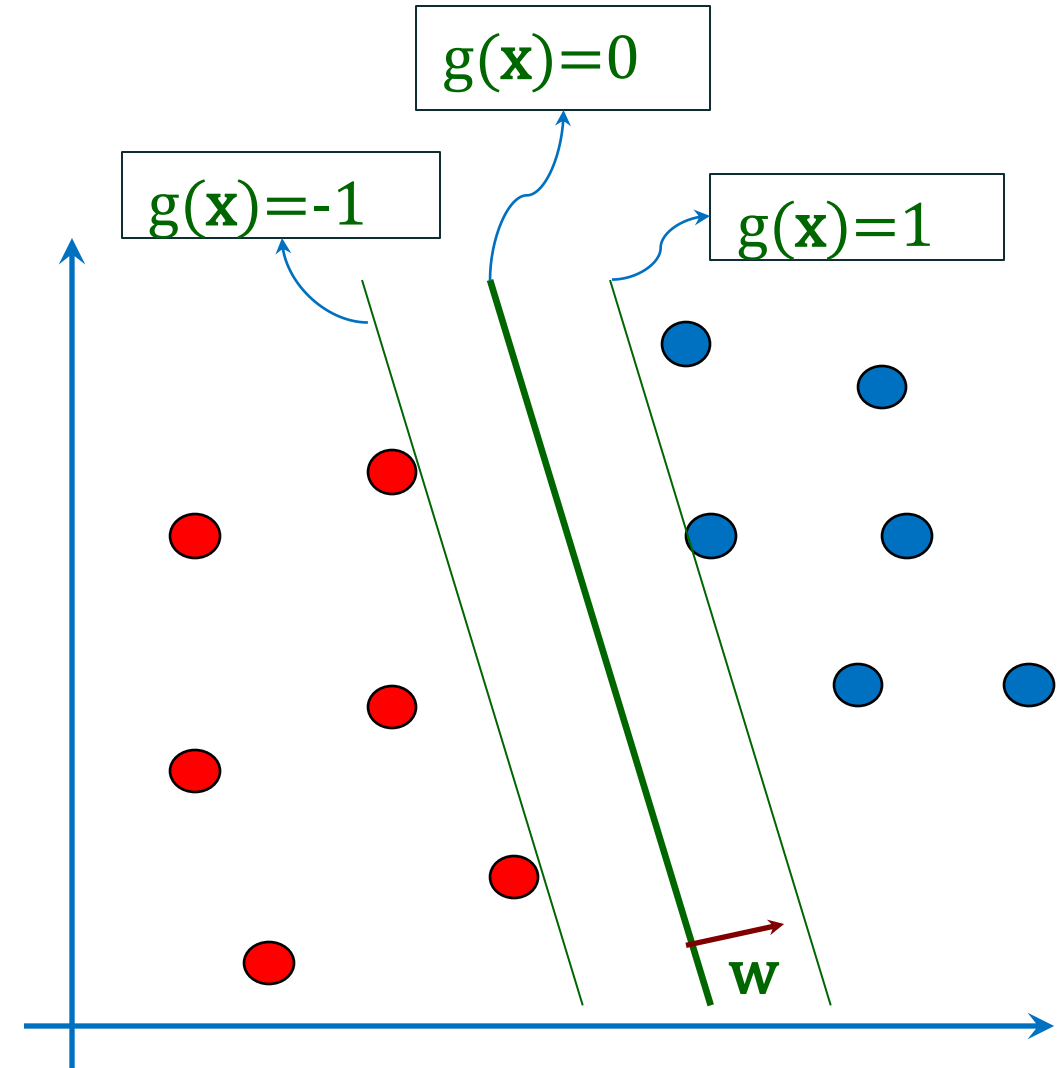
SVM: Formulation

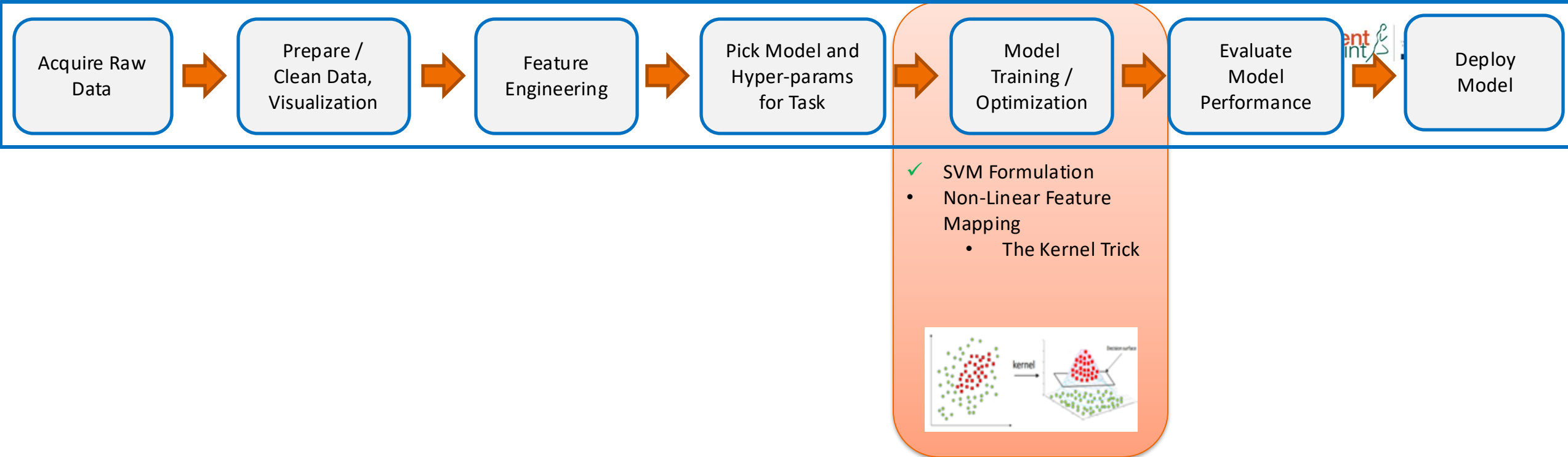
- Let $g(X) = W^T X + b$
- We want to maximize margin:
 - $W^T X_i + b \leq -1$ for $y_i = -1$
 - $W^T X_i + b \geq 1$ for $y_i = 1$
 - Or $y_i(W^T X_i + b) \geq 1$ for i .



SVM: Formulation

- Mathematically, Minimize
 $\frac{1}{2}W^TW$
- Subject to:
 $-y_i(W^TX_i + b) \geq 1$ for all i .
- This is convex optimization.
- Exact solutions exist



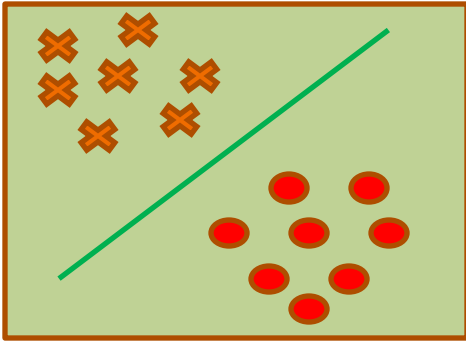


SVMs and Kernels

Kernel as Similarity Function

“Linear” Learning techniques

- Linear classifier



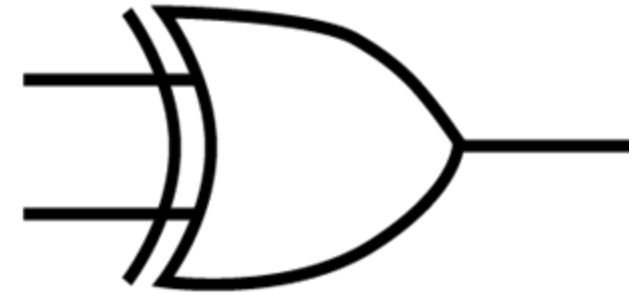
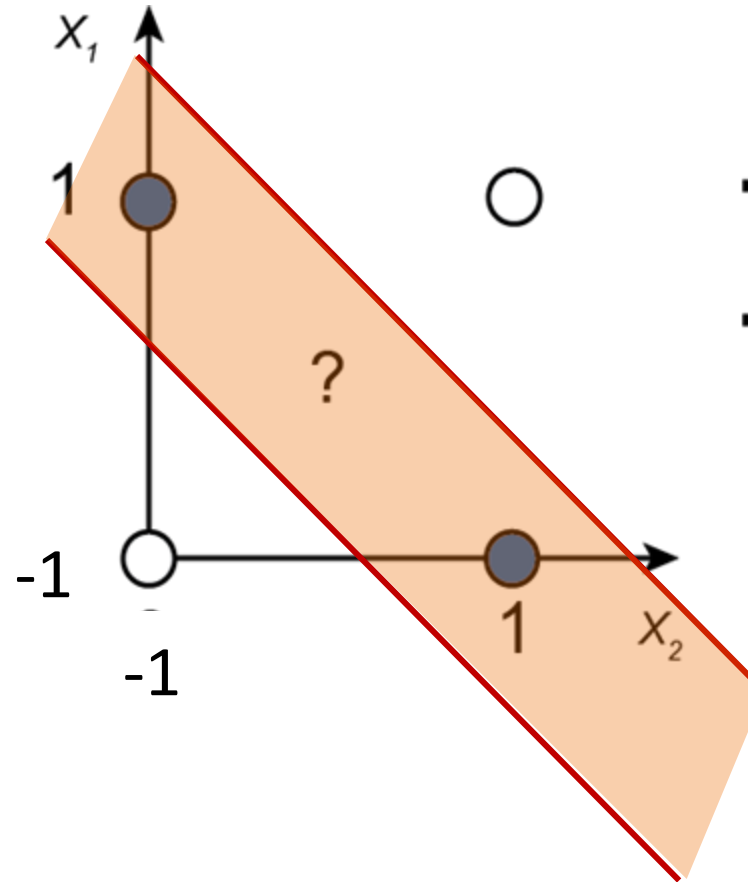
$$g(x_n) = \text{sign}(w^T x_n)$$

where w is an d -dim vector (learned)

- Techniques:
 - Perceptron
 - Logistic regression
 - Support vector machine (SVM)
 - Etc.

XOR: Limitation of Linear Methods

x_1	x_2	$x_1 \text{ XOR } x_2$
-1	-1	-1 (-)
-1	1	1 (+)
1	-1	1 (+)
1	1	-1 (-)



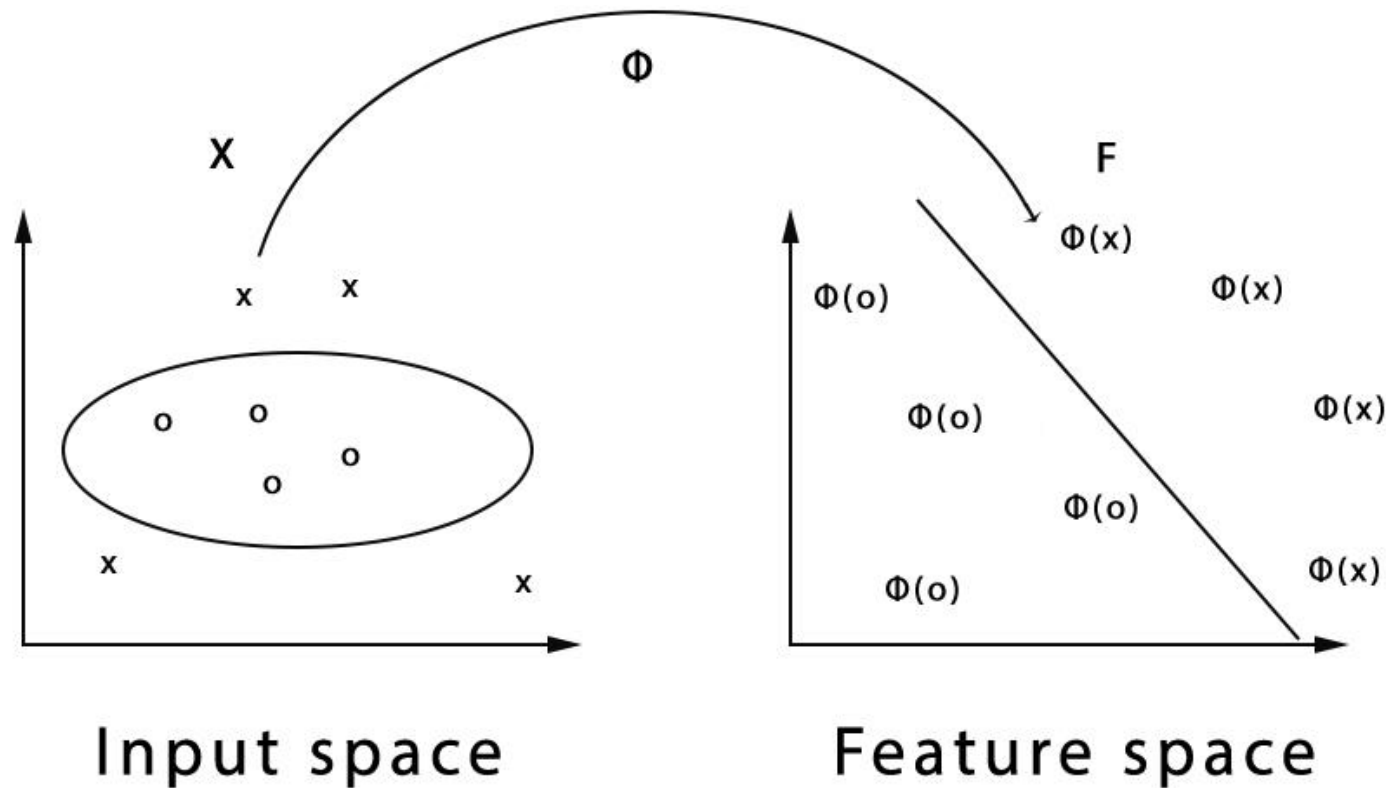
Consider a new feature $x_3 = x_1x_2$

- With a “new feature”, a problem that is linearly non-separable has become separable!!
- A difficult problem became easy!!

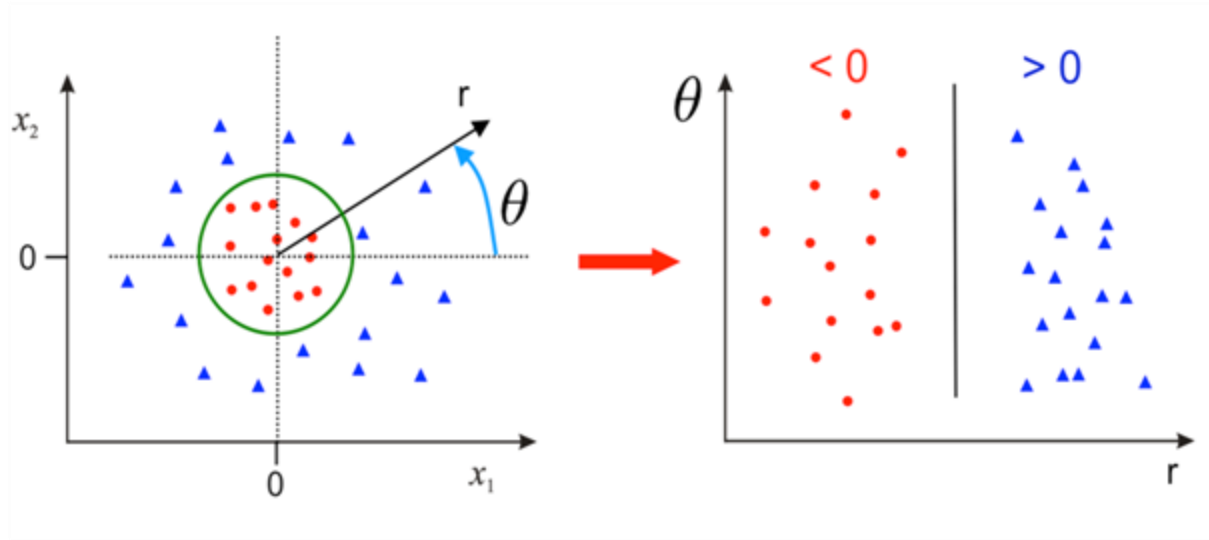
X1	X2	X3	XOR
-1	-1	1	-
-1	1	-1	+
1	-1	-1	+
1	1	1	-

Non Linear SVM

- Some data points are not linear separable
- **Intuition:** to transform the data to a high dimension space



Nonlinearity with Feature Maps

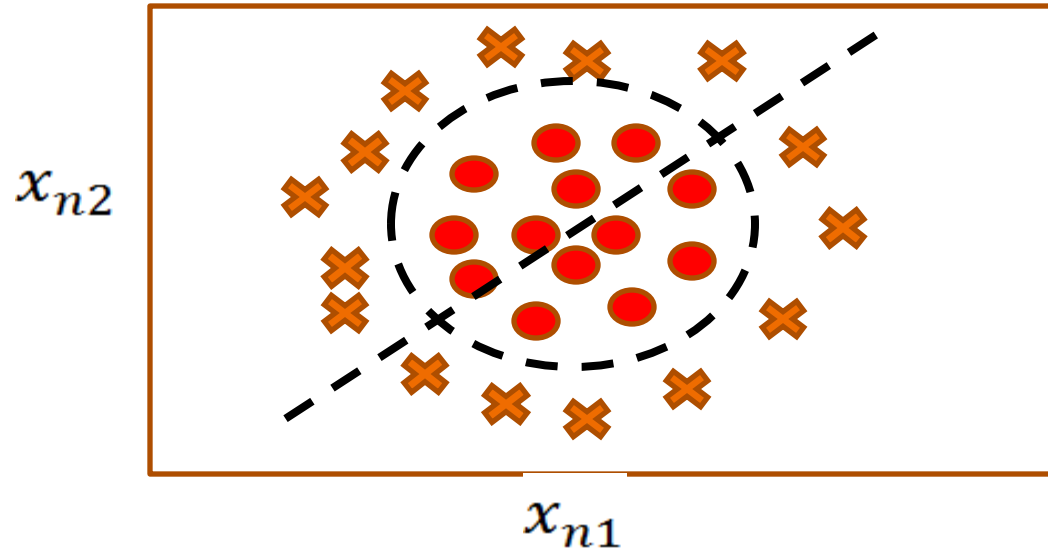


$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \rightarrow \begin{pmatrix} r \\ \theta \end{pmatrix}$$

- With a “smart” feature map, a linearly non-separable problem can be converted to a separable problem.!!
- The feature mapping is often denoted by: $\phi(X)$

More General

- Non-linear case



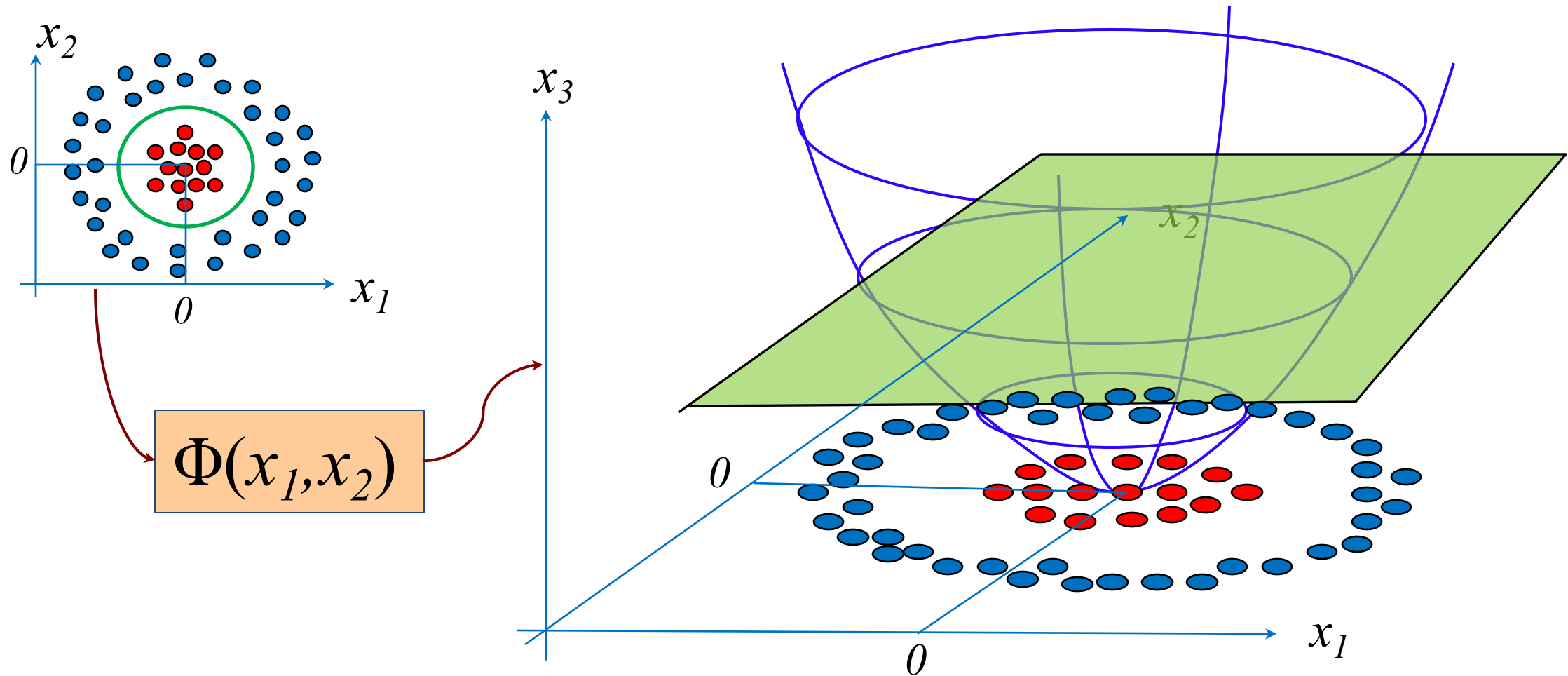
$$x_n = [x_{n1}, x_{n2}]$$



$$x_n = [x_{n1}, x_{n2}, x_{n1} * x_{n2}, x_{n1}^2, x_{n2}^2]$$

$$g(x_n) = \text{sign}(w^T x_n)$$

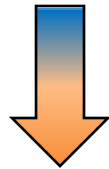
Non-linear Mapping



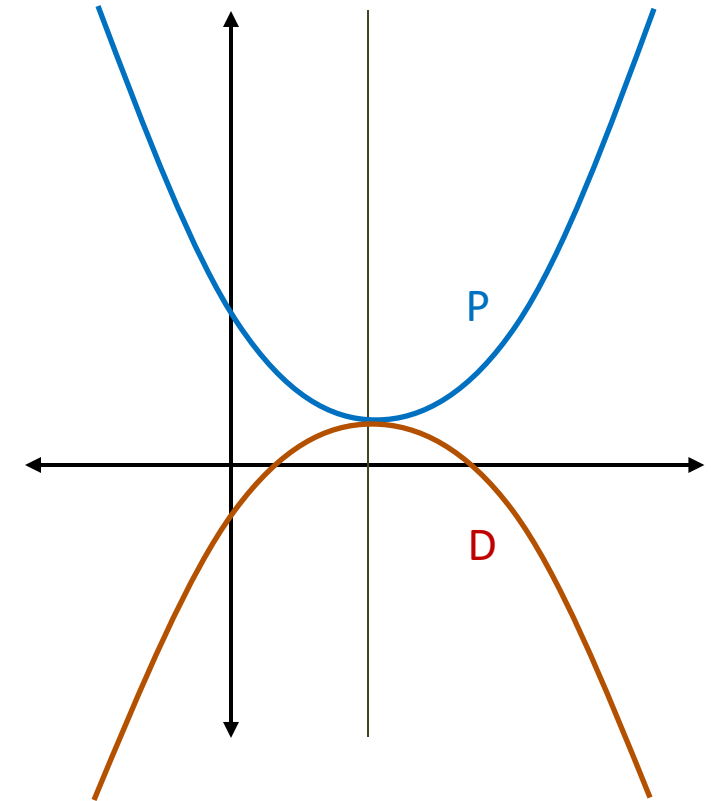
Φ is a non-linear mapping into a possibly high-dimensional space

SVM: Primal and Dual

$$\begin{aligned} \text{Minimize: } \phi(w) &= \frac{1}{2} w^T w \\ \text{Subject to: } y_i(w^T x_i + b) - 1 &\geq 0 \quad \forall i \end{aligned}$$



$$\begin{aligned} \text{Maximize: } Q(\alpha) &= \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j x_i^T x_j \\ \text{Subject to: } \alpha_i &\geq 0 \quad \forall i \text{ and } \sum_{i=1}^N \alpha_i y_i = 0 \end{aligned}$$



$$w_0 = \sum_{i=1}^N \alpha_i y_i x_i$$

Kernel Strategy

$$w = \sum_{i=1}^N \alpha_i y_i x_i$$

What we need is only $w^T x = \sum_{i=1}^N \alpha_i y_i x_i^T x$

We can do the same in a new feature space:

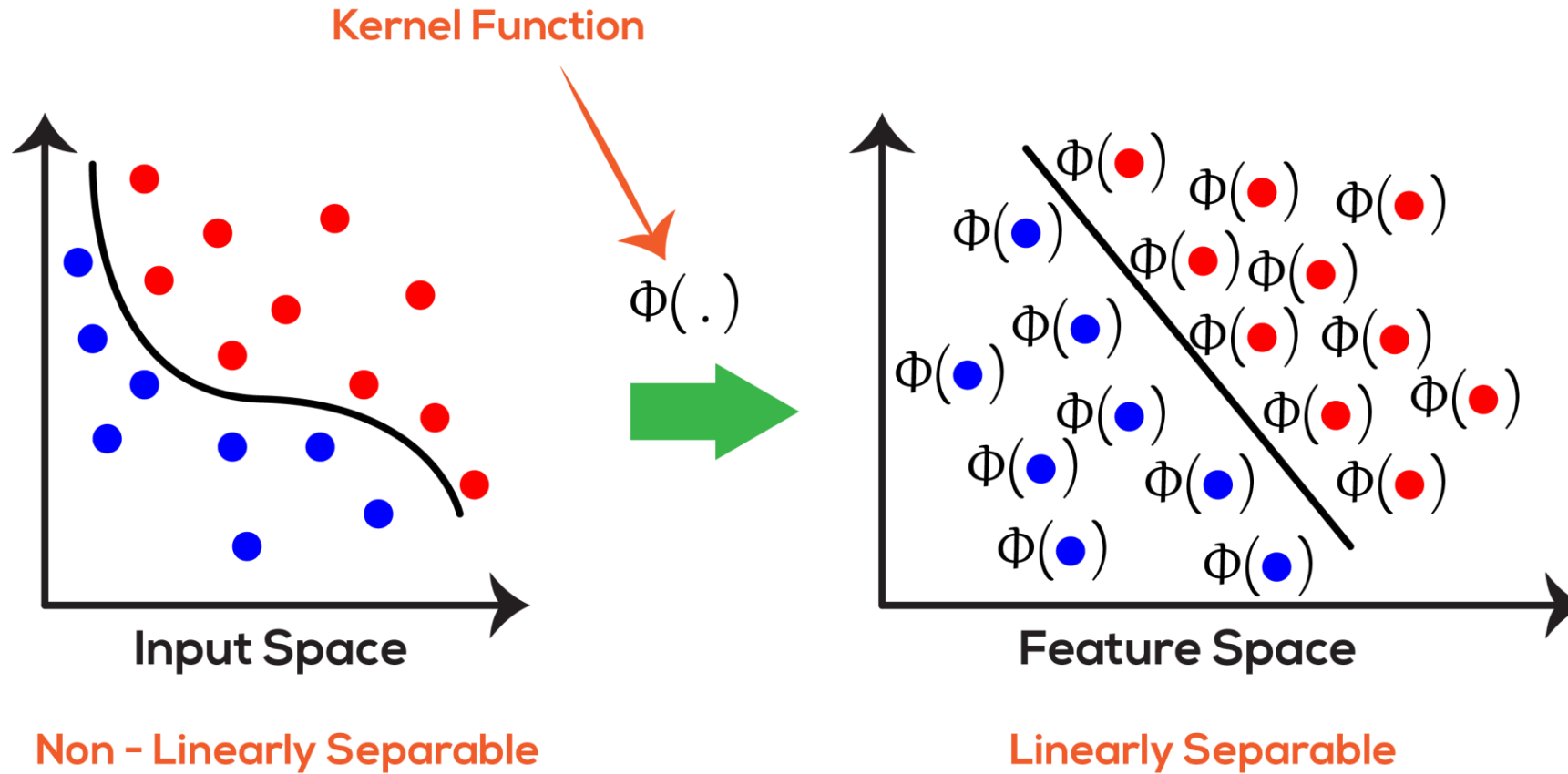
$$w^T x = \sum_{i=1}^N \alpha_i y_i \phi(x_i)^T \phi(x) \qquad w^T x = \sum_{i=1}^N \alpha_i y_i K(x_i, x)$$

Kernels

- Interestingly, it is possible to do this without explicitly doing the non-linear mapping to high dimensions $K(x_i, x_j)$
- We need only a kernel function $K(s_i, x_i) = \phi(s_i) \cdot \phi(x_i)$

Non Linear SVM

- Transform the data to a high dimension space



Popular Kernels

- Polynomial:

$$K_p(X_1, X_2) = (1 + X_1 \cdot X_2)^p$$

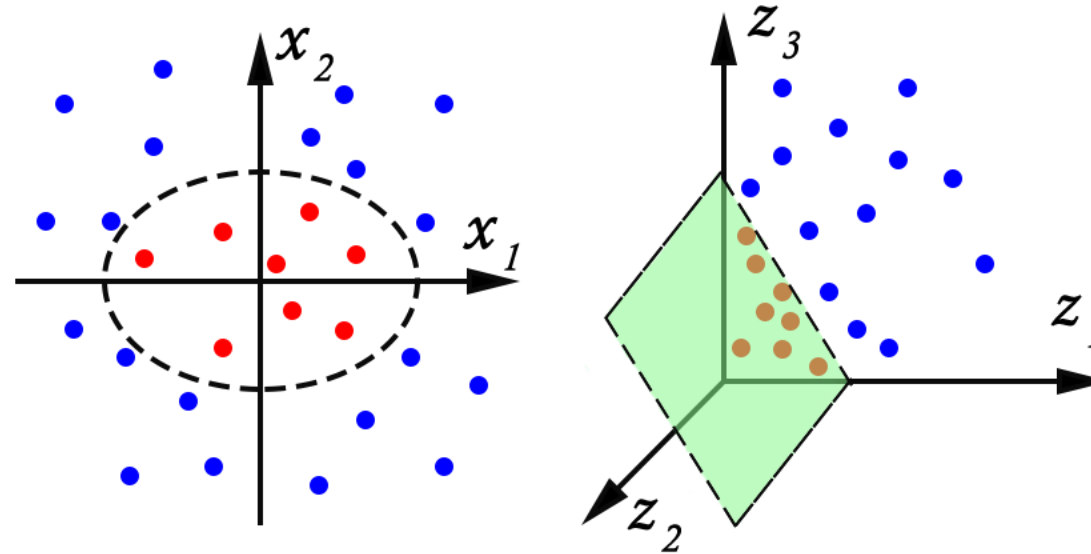
- Radial Basis Function (RBF) or Gaussian:

$$K_r(X_1, X_2) = e^{-\frac{(\|X_1 - X_2\|)^2}{2\sigma^2}}$$

- Hyperbolic Tangent:

$$K_s(X_1, X_2) = \tanh(\beta_0 X_1 \cdot X_2 + \beta_1)$$

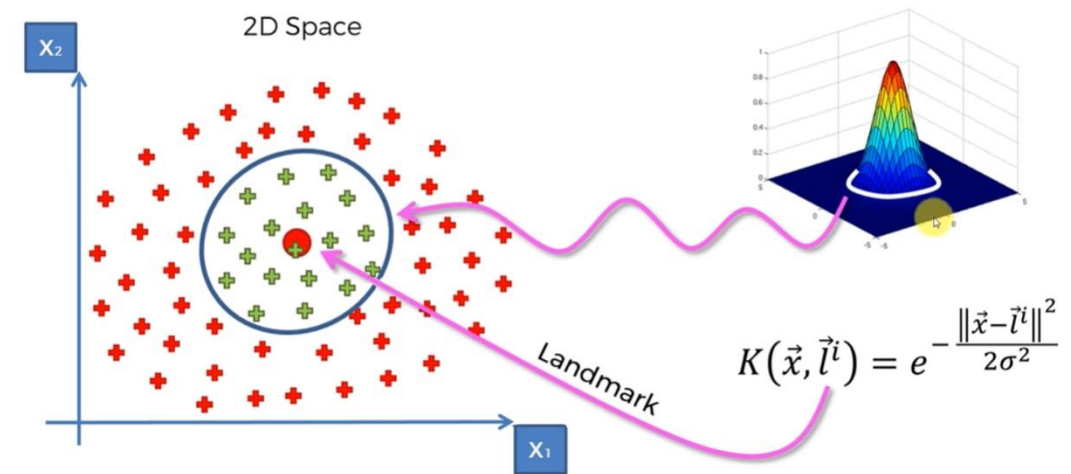
Polynomial Mapping



- $\phi : R^2 \rightarrow R^3$
- $(X_1, X_2) \rightarrow (Z_1, Z_2, Z_3) = (X_1^2, \sqrt{(2)}X_1X_2, X_2^2)$
- $K_p(X_1, X_2) = (1 + X_1 \cdot X_2)^p$

Radial Basis Function (RBF)

- $K_r(\vec{x}, \vec{l}^i) = e^{-\frac{(\|\vec{x} - \vec{l}^i\|)^2}{2\sigma^2}}$
- Where,
 - $\vec{x}$: X vector (some point in our dataset)
 - $\vec{l}^i$: l stands for landmarks (i represents the number of landmarks)
 - $(\|\vec{x} - \vec{l}^i\|)^2$: Distance between a point and the landmark squared



Summary

- Linear SVMs generalize well, but cannot separate non-linear data
- If features can be transformed appropriately, simple linear algorithms (classification, regression) are enough.
- How do we find the feature transformation?
 - Make a reasonable guess?
 - Ans: Use some popular complex functions.
- Kernels (nonlinear) SVMs are also good at generalization and can deal with non-linear data.
- Need not be as efficient/compact.

Reference Links

- https://www.youtube.com/watch?v=9_DJ4KvyYoo

Demo Experiments

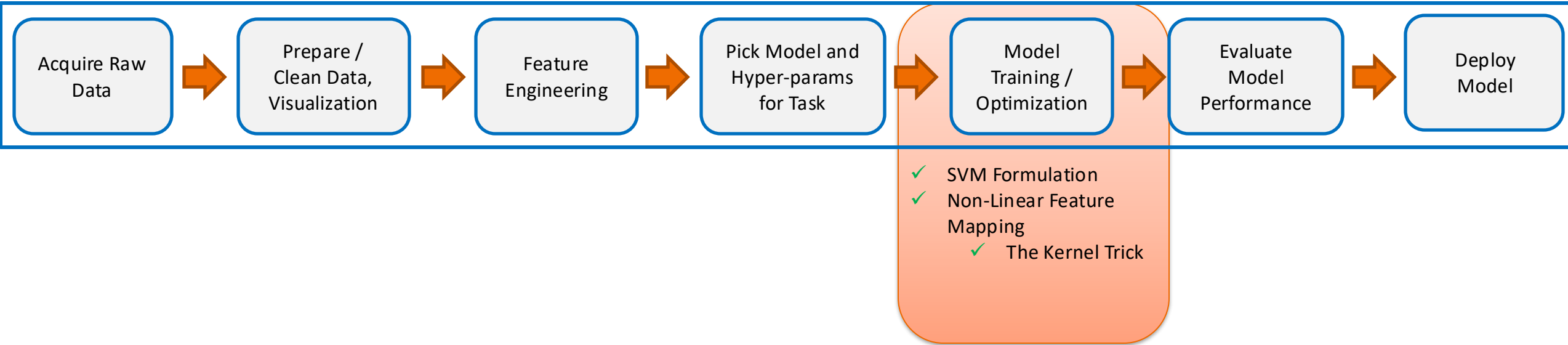
Demo_Non_Linear_SVM_multi-class

The aim of this experiment is to perform binary and multi-class classification on Iris dataset and get the support vectors that are the basis for the max margin.

Demo_Non-Linear_SVM_RBF

The aim of this experiment is to understand how non-linear separable data can be visualized linearly in a higher dimensional space

Summary



Thanks!!

Questions?